

Homework 9: Linear Recurrence Relations – Solving via Matrix Techniques

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Introduction.

One of the most useful aspects of diagonalization and matrix functions is their ability to swiftly solve problems that do not even seem to be linear algebraic in nature at first glance. We have already seen a hint of how this works in *homework 8* where we developed a closed-form formula for the Fibonacci numbers. In this homework, we extend this idea to a larger family of sequences of numbers $x_1, x_2, x_3, \dots$ that are defined in a similar linear way.

A linear recurrence relation is a formula that represents a term x_n in a sequence of numbers as a linear combination of the immediately previous terms in the sequence. That is, it is a formula for which there exist scalars $a_0, a_1, \dots, a_{k-1}$ so that

$$x_n = a_{k-1}x_{n-1} + a_{k-2}x_{n-2} + \dots + a_0x_{n-k}, \quad \forall, \quad n \geq k.$$

The number of terms in the linear combination is called the degree of the recurrence relation (so the recurrence relation has degree k). In order for the sequence to be completely defined, we also need some initial conditions that specify the first term(s) of the sequence. In particular, for a degree- k recurrence relation we require the first k terms to be specified.

The characteristic polynomial of the degree- k linear recurrence relation

$$x_n = a_{k-1}x_{n-1} + a_{k-2}x_{n-2} + \dots + a_0x_{n-k}$$

is the degree- k polynomial

$$p(\lambda) = \lambda^k - a_{k-1}\lambda^{k-1} - a_{k-2}\lambda^{k-2} - \dots - a_1\lambda - a_0.$$

Problem. Find an explicit formula for the n -th term of the sequence defined by the recurrence relation

$$x_n = 8x_{n-1} - 17x_{n-2} + 10x_{n-3}, \quad n \geq 3,$$

if $x_0 = 1, x_1 = 4$, and $x_2 = 22$.

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Good Luck